

CS4226

Internet Architecture

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School of Computing

National University of Singapore

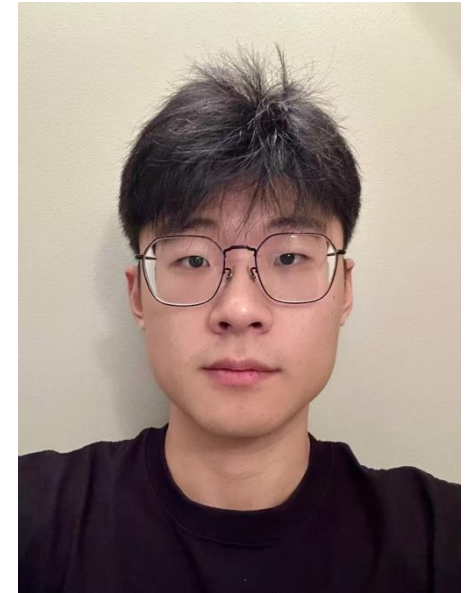
About Your Lecturer



- Email: *tbma@comp.nus.edu.sg*
- Office: COM3 #02-14
- Website: *www.comp.nus.edu.sg/~tbma/*
- Research areas:
 - ❖ Computer Networks
 - ❖ Distributed Systems

About Your TA

- ❑ Tiantong HU
- ❑ Email: *tiantong.hu@u.nus.edu*
- ❑ Office: Systems and Networking Research Lab at COM3
- ❑ Research: Software-Defined Networking



About Your TA

- Jianzhi Shi
- Email: *e1583353@u.nus.edu*
- Office: Systems and Networking Research Lab at COM3
- Research: Network Security



A Survey

- ☐ How many of you are non-CS students?
- ☐ How many of you are 2nd year undergraduate students?
- ☐ How many of you are 3rd year undergraduate students?
- ☐ How many of you are 4th year undergraduate students?

About this course

- ❑ Continuation of CS2105
- ❑ In parallel with CS3103
- ❑ For focus area Networking and Distributed Systems (was called Computer Networks)
- ❑ Cover more advanced topics on
 - ❖ Network architectures
 - ❖ Network management
 - ❖ Networks performance

<https://nusmods.com/courses/CS4226/internet-architecture>

The old focus area: Computer Networks

**CS2105 Introduction to
Computer Networks**

**CS3103L Computer
Networks Laboratory (2MC)**

**CS3103 Computer
Networks and Protocols**

**CS4274 Mobile and
Multimedia Networking**

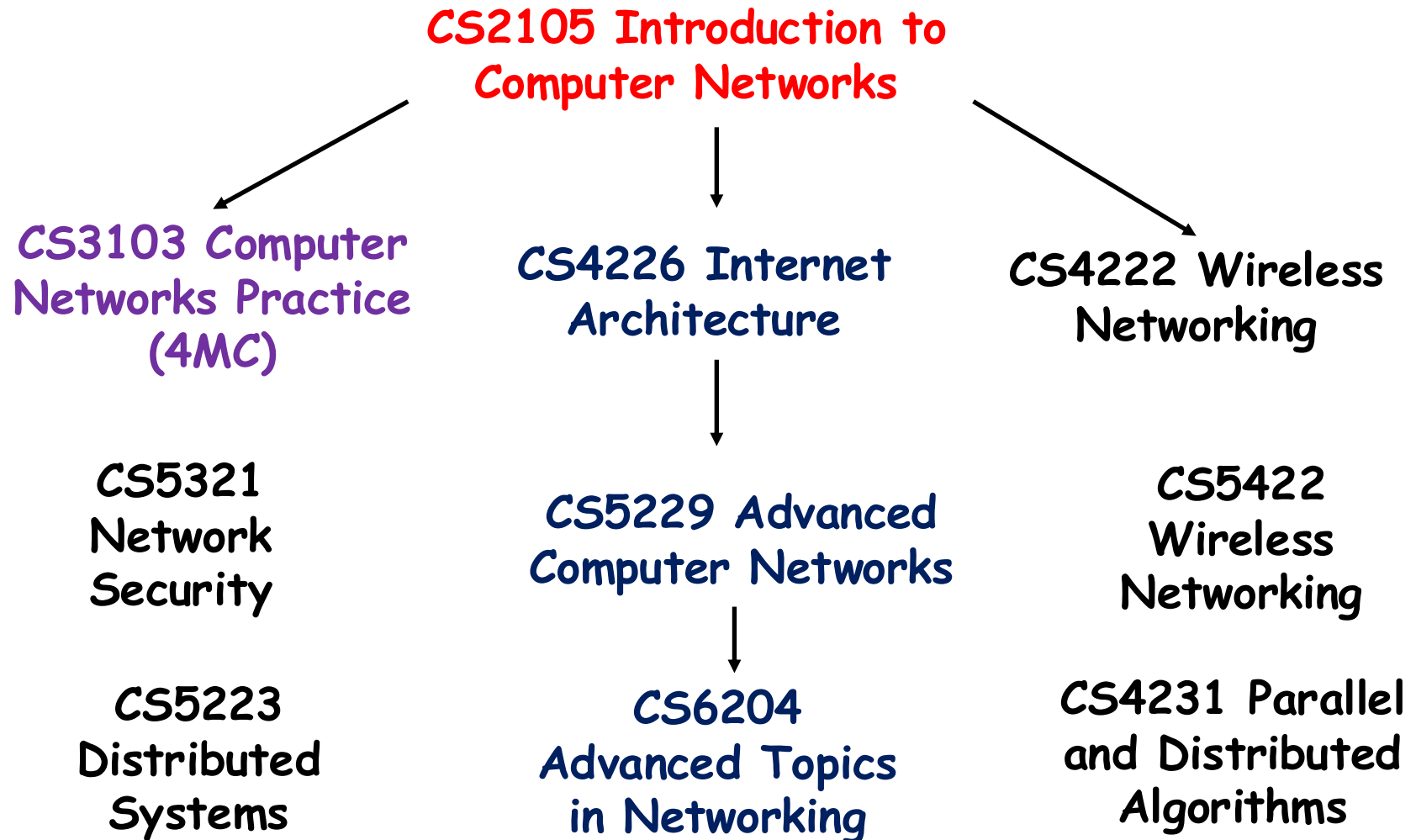
**CS4222 Wireless Computing
and Sensor Networks**

**CS4344 Networked
and Mobile Gaming**

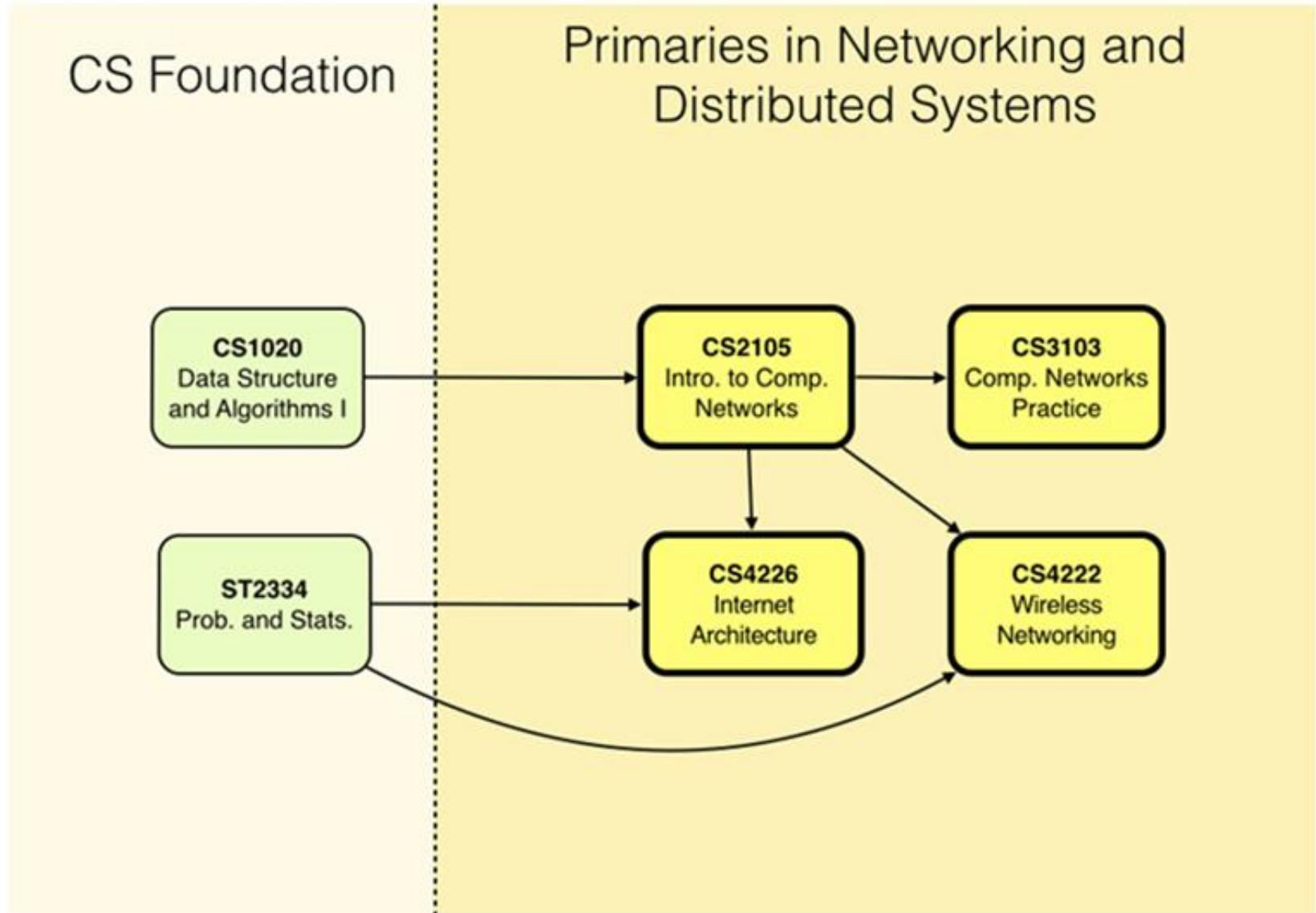
**CS5229 Advanced
Computer Networks**

**CS5248 Systems Support
for Continuous Media**

Networking and Distributed Systems



Networking and Distributed Systems



Pre-requisites from CS2105

CS2105 Lecture 1

Introduction

14 January, 2014

After this class, you are expected to:

- understand the basic terms, including host, packet, protocol, throughput, bottleneck link, store-and-forward, and autonomous system.
- know about the logical (the five layers) and physical architecture (as a network of ASes) of the Internet.
- know about the pros and cons of packet switching versus circuit switching.
- understand the different components of end-to-end delay and their relations to bandwidth, packet size, distance, propagation speed, and queue size.

Background Reference Book

Computer Networking: A Top-Down Approach: International Edition, 6/e

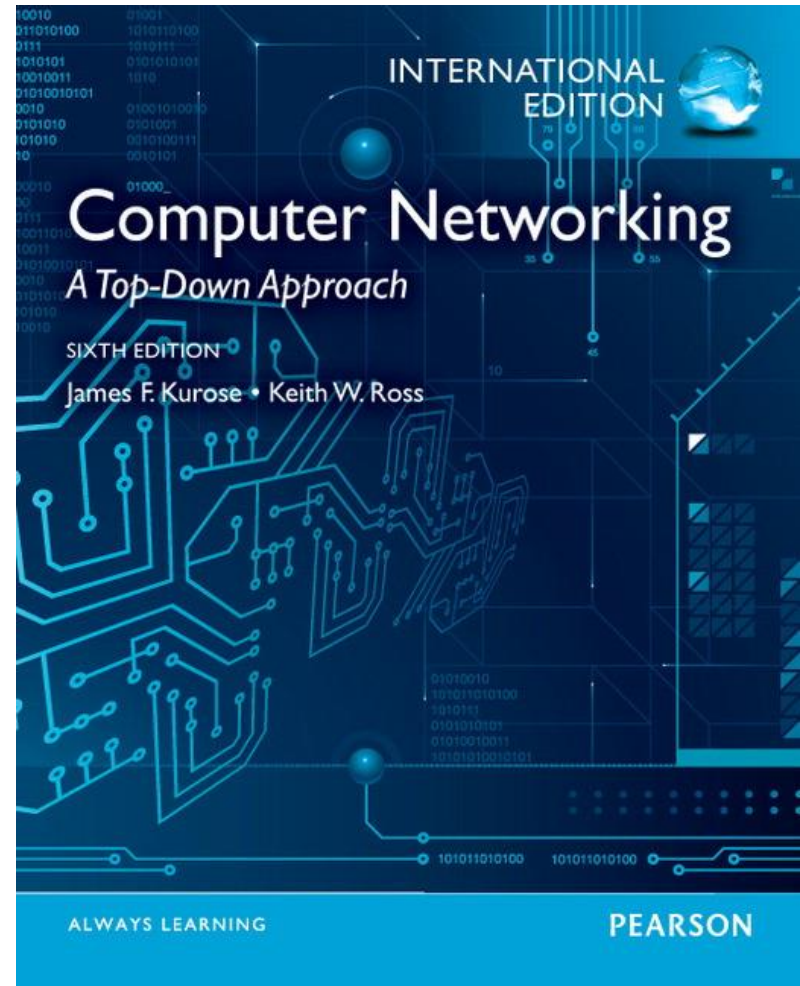
Author : Kurose

Ross

Publisher : Pearson

ISBN : 9780273768968

Acknowledgement:
Some lecture slides will be
taken from this book!



CS3103: Computer Networks Practice

□ Topics covered by CS3103

- ❖ IPv4, NAT, ICMP, ARP, DHCP and DNS
- ❖ Design and Configure VLAN
- ❖ Wireless Protocol (802.11b/g/n)
- ❖ Intra-Domain Routing in Detail - RIP and OSPF
- ❖ UDP and TCP (exp. in-depth congestion control)
- ❖ Security Fundamentals, IP-Sec-VPN, SSL, PGP
- ❖ Multimedia Basics, RTP, RTSP, VoIP with SIP
- ❖ Traffic engineering with MPLS
- ❖ Configuring/Programming open switch (SDN)
- ❖ App. Layer Protocols and Socket Programming

CS3103 Vs. CS4226

□ CS3103 Computer Networks Practice

- ❖ network protocols
- ❖ vertical view (protocol stack):
 - data link, network and transport layer protocols
- ❖ more practical/hands-on experiences

□ CS4226 Internet Architecture

- ❖ network architecture and network management
- ❖ horizontal view (architectural design):
 - resource allocation, inter-domain routing, peer to peer networks and economics of the ecosystem
- ❖ design principles, model, theory and analysis

Tentative Topics of CS4226

- ❑ Network Architectures
 - ❖ Peer-to-Peer Networks
- ❑ Internet Ecosystem
 - ❖ Inter-Domain routing: BGP
 - ❖ Business and Economics
- ❑ Network Management Paradigm
 - ❖ Software-Defined Networking
- ❑ Network Performance
 - ❖ Queueing model and analysis
 - ❖ Resource allocation

Tentative Plan

- Aug 14 Logistics, Performance Metrics
- Aug 21 Probability and Queueing Model
- Aug 28 M/M/1 Model and Analysis
- Sep 4 Queueing Networks
- Sep 11 Network Resource Allocation
- Sep 18 Introduction to SDN
- Sep 25 Recess
- Oct 2 Mid-term Exam (MPSH)

Tentative Plan

- ❑ Oct 9 Software Defined Networking
- ❑ Oct 16 Inter-Domain Routing Protocol
- ❑ Oct 23 BGP Routing Policy
- ❑ Oct 30 Internet Exchange Point
- ❑ Nov 6 **Conference Travel**
- ❑ Nov 13 Peer-to-Peer Networks
- ❑ Dec 1 **Final Exam**

Tentative Course Assessment

- ❑ Written assignment 10% -> 5%
- ❑ Programming assignments 35% -> 40%
- ❑ Mid-term Exam 20%
- ❑ Final Exam 35%
- ❖ Workload 2-1-0-4-3 (see from NUSMODS)
 - ❖ lecture + tutorial + lab hours, and
 - ❖ no. of hours for projects, assignments, etc + hours for preparatory work

Mode of Teaching

- Lectures and tutorials will also be
 - ❖ recorded and posted via Canvas
 - ❖ no real-time Zoom sessions
- Midterm/final exams will be
 - ❖ physical face-to-face tests
 - ❖ using online tool i.e., Exemplify
- Feel free to contact me and the TA
 - ❖ raise questions in class, send emails after class
 - ❖ arrange individual meetings upon requests

Tutorial Time and Style

- ❑ Tutorial time: starting from Week 3
 - ❖ Due to the scale of the class, we will host a single tutorial session after lecture in LT15
 - ❖ Similar to lectures, we will record tutorials
- ❑ Tutorial material/questions will be released one week ahead
- ❑ Do the tutorial problems yourself!
 - ❖ View the recorded tutorials
 - ❖ Ask TA or me about it if you still quite get it

Discussion Forum: Piazza

- ❑ We'll use Piazza as our discussion forum
 - ❖ please sign up at:
<https://piazza.com/nus.edu.sg/fall2026/cs4226>
- ❑ Currently four folders created
 - ❖ lecture, tutorial, assignment and exam
- ❑ We'll still use Canvas for distributing
 - ❖ announcements
 - ❖ lecture/tutorial slides
 - ❖ assignments
 - ❖ recorded videos

Duration of Classes

- University Policy on the Duration of Classes:
 - ❖ For 2-hour lessons, the break should be kept 5 minutes, and students released 25 minutes before the end of the second hour.
- We will try to take a 5-minute break in between, and target to complete the lecture ~15 minutes before the end of the 2nd hour.